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Methods for forming three-dimensional memory device

Abstract

Embodiments of three-dimensional (3D) memory devices and methods for forming the same are disclosed. In an example, a method for forming a 3D memory device comprises forming a channel structure extending vertically through a memory stack into a semiconductor layer on a substrate. The memory stack comprises interleaved stack conductive layers and stack dielectric layers. The method further comprises forming an insulating structure in an opening extending vertically through the memory stack and at a distance away from the channel structure, and comprising a dielectric layer doped with at least one of hydrogen or an isotope of hydrogen.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. application Ser. No. 17/020,457, filed on Sep. 14, 2020, entitled “THREE-DIMENSIONAL MEMORY DEVICE WITH HYDROGEN-RICH SEMICONDUCTOR CHANNELS,” which is a continuation of International Application No. PCT/CN2020/105686, filed on Jul. 30, 2020, entitled “THREE-DIMENSIONAL MEMORY DEVICE WITH HYDROGEN-RICH SEMICONDUCTOR CHANNELS,” both of which are incorporated herein by reference in their entireties.

BACKGROUND

(1) Embodiments of the present disclosure relate to three-dimensional (3D) memory devices and fabrication methods thereof.

(2) Planar memory cells are scaled to smaller sizes by improving process technology, circuit design, programming algorithm, and fabrication process. However, as feature sizes of the memory cells approach a lower limit, planar process and fabrication techniques become challenging and

costly. As a result, memory density for planar memory cells approaches an upper limit.

(3) A 3D memory architecture can address the density limitation in planar memory cells. The 3D memory architecture includes a memory array and peripheral devices for controlling signals to and from the memory array.

SUMMARY

(4) Embodiments of 3D memory devices and methods for forming the same are disclosed herein.

(5) In one example, a 3D memory device includes a memory stack including interleaved stack conductive layers and stack dielectric layers, a semiconductor layer, a plurality of channel structures each extending vertically through the memory stack into the semiconductor layer, and an insulating structure extending vertically through the memory stack and including a dielectric layer doped with at least one of hydrogen or an isotope of hydrogen.

(6) In another example, a 3D memory device includes a memory stack including interleaved stack conductive layers and stack dielectric layers, a semiconductor layer, a plurality of channel structures each extending vertically through the memory stack into the semiconductor layer, and an insulating structure extending vertically through the memory stack and including a dielectric layer. Each of the channel structures includes a memory film and a semiconductor channel doped with at least one of hydrogen or an isotope of hydrogen.

(7) In still another example, a method for forming a 3D memory device is disclosed. A channel structure extending vertically through a memory stack into a semiconductor layer on a substrate is formed. The memory stack includes interleaved stack conductive layers and stack dielectric layers. An insulating structure in an opening extending vertically through the memory stack and including a dielectric layer doped with at least one of hydrogen or an isotope of hydrogen is formed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated herein and form a part of the specification, illustrate embodiments of the present disclosure and, together with the description, further serve to explain the principles of the present disclosure and to enable a person skilled in the pertinent art to make and use the present disclosure.

(2) FIG. 1A illustrates a side view of a cross-section of an exemplary 3D memory device with hydrogen-rich semiconductor channels, according to some embodiments of the present disclosure.

(3) FIG. 1B illustrates a side view of a cross-section of another exemplary 3D memory device with hydrogen-rich semiconductor channels, according to some embodiments of the present disclosure.

(4) FIGS. 2A-2H illustrate side views of cross-sections of various exemplary hydrogen-rich insulating structures in the 3D memory device in FIG. 1A or 1B, according to various embodiments of the present disclosure.

(5) FIGS. 3A-3M illustrate a fabrication process for forming an exemplary 3D memory device with hydrogen-rich semiconductor channels, according to some embodiments of the present disclosure.

(6) FIG. 4 illustrates a flowchart of a method for forming an exemplary 3D memory device with hydrogen-rich semiconductor channels, according to some embodiments of the present disclosure.

(7) Embodiments of the present disclosure will be described with reference to the accompanying drawings.

DETAILED DESCRIPTION

(8) Although specific configurations and arrangements are discussed, it should be understood that this is done for illustrative purposes only. A person skilled in the pertinent art will recognize that other configurations and arrangements can be used without departing from the spirit and scope of the present disclosure. It will be apparent to a person skilled in the pertinent art that the present disclosure can also be employed in a variety of other applications.

(9) It is noted that references in the specification to “one embodiment,” “an embodiment,” “an example embodiment,” “some embodiments,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases do not necessarily refer to the same embodiment. Further, when a particular feature, structure or characteristic is described in connection with an embodiment, it would be within the knowledge of a person skilled in the pertinent art to effect such feature, structure or characteristic in connection with other embodiments whether or not explicitly described.

(10) In general, terminology may be understood at least in part from usage in context. For example, the term “one or more” as used herein, depending at least in part upon context, may be used to describe any feature, structure, or characteristic in a singular sense or may be used to describe combinations of features, structures or characteristics in a plural sense. Similarly, terms, such as “a,” “an,” or “the,” again, may be understood to convey a singular usage or to convey a plural usage, depending at least in part upon context. In addition, the term “based on” may be understood as not necessarily intended to convey an exclusive set of factors and may, instead, allow for existence of additional factors not necessarily expressly described, again, depending at least in part on context.

(11) It should be readily understood that the meaning of “on,” “above,” and “over” in the present disclosure should be interpreted in the broadest manner such that “on” not only means “directly on” something but also includes the meaning of “on” something with an intermediate feature or a layer therebetween, and that “above” or “over” not only means the meaning of “above” or “over” something but can also include the meaning it is “above” or “over” something with no intermediate feature or layer therebetween (i.e., directly on something).

(12) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

(13) As used herein, the term “substrate” refers to a material onto which subsequent material layers are added. The substrate itself can be patterned. Materials added on top of the substrate can be patterned or can remain unpatterned. Furthermore, the substrate can include a wide array of semiconductor materials, such as silicon, germanium, gallium arsenide, indium phosphide, etc. Alternatively, the substrate can be made from an electrically non-conductive material, such as a glass, a plastic, or a sapphire wafer.

(14) As used herein, the term “layer” refers to a material portion including a region with a thickness. A layer can extend over the entirety of an underlying or overlying structure or may have an extent less than the extent of an underlying or overlying structure. Further, a layer can be a region of a homogeneous or inhomogeneous continuous structure that has a thickness less than the thickness of the continuous structure. For example, a layer can be located between any pair of horizontal planes between, or at, a top surface and a bottom surface of the continuous structure. A layer can extend horizontally, vertically, and/or along a tapered surface. A substrate can be a layer, can include one or more layers therein, and/or can have one or more layer thereupon, thereabove, and/or therebelow. A layer can include multiple layers. For example, an interconnect layer can include one or more conductor and contact layers (in which interconnect lines and/or vertical interconnect access (via) contacts are formed) and one or more dielectric layers.

(15) As used herein, the term “nominal/nominally” refers to a desired, or target, value of a characteristic or parameter for a component or a process operation, set during the design phase of a product or a process, together with a range of values above and/or below the desired value. The

range of values can be due to slight variations in manufacturing processes or tolerances. As used herein, the term “about” indicates the value of a given quantity that can vary based on a particular technology node associated with the subject semiconductor device. Based on the particular technology node, the term “about” can indicate a value of a given quantity that varies within, for example, 10-30% of the value (e.g., $\pm 10\%$, $\pm 20\%$, or $\pm 30\%$ of the value).

(16) As used herein, the term “3D memory device” refers to a semiconductor device with vertically oriented strings of memory cell transistors (referred to herein as “memory strings,” such as NAND memory strings) on a laterally-oriented substrate so that the memory strings extend in the vertical direction with respect to the substrate. As used herein, the term “vertical/vertically” means nominally perpendicular to the lateral surface of a substrate.

(17) In fabricating some 3D memory devices, more thin films (e.g., silicon oxide, silicon nitride, polysilicon, etc.) need to be deposited on the silicon substrate as the level of memory cells increases. For example, in 3D NAND memory devices, more thin films need to fill in the staircase area, core array area, and gate line slits, which may cause thin film deformation when the thin film structures become more complex, in particular, after thermal processes. The stress due to thin film deformation can further cause the deformation of the entire wafer (e.g., wafer bow), thereby reducing the production yield.

(18) Moreover, 3D NAND memory devices usually use polysilicon as the material of the semiconductor channels. Polysilicon, however, may have defects between grain boundaries and at interfaces (e.g., between semiconductor channel and tunneling layer), which can affect the memory cell performance. For example, dangling bonds can be found at the interfaces, which can attract carriers to reduce the carrier mobility in the semiconductor channels. The dangling bonds may also affect the grain size of polysilicon to increase the internal stress of the polysilicon semiconductor channels.

(19) Various embodiments in accordance with the present disclosure provide 3D memory devices with hydrogen-rich semiconductor channels to improve the cell performance and adjust the wafer stress. Hydrogen and its isotopes, such as protium, deuterium, or tritium, can form stable H—Si covalent bonds to reduce the number of dangling bonds in the polysilicon semiconductor channels, thereby improving the carrier mobility and adjusting the internal stress. In some embodiments, 3D memory devices include hydrogen-rich insulating structures, for example, by depositing dielectric layers (e.g., silicon nitride) doped with at least one of hydrogen or an isotope of hydrogen in the gate line slits. During the later thermal processes, the hydrogen or its isotope can diffuse from the hydrogen-rich insulating structure to the polysilicon semiconductor channels to form hydrogen-rich semiconductor channels, thereby curing the defects of the polysilicon and adjusting the wafer stress. As a result, memory cell performance can be improved, and the production yield can be increased.

(20) FIG. 1A illustrates a side view of a cross-section of an exemplary 3D memory device **100** with hydrogen-rich semiconductor channels, according to some embodiments of the present disclosure. In some embodiments, 3D memory device **100** is a bonded chip including a first semiconductor structure **102** and a second semiconductor structure **104** stacked over first semiconductor structure **102**. First and second semiconductor structures **102** and **104** are joined at a bonding interface **106** therebetween, according to some embodiments. As shown in FIG. 1A, first semiconductor structure **102** can include a substrate **101**, which can include silicon (e.g., single crystalline silicon, c-Si), silicon germanium (SiGe), gallium arsenide (GaAs), germanium (Ge), silicon on insulator (SOI), or any other suitable materials.

(21) First semiconductor structure **102** of 3D memory device **100** can include peripheral circuits **108** on substrate **101**. It is noted that x-, y-, and z-axes are included in FIG. 1A to illustrate the spatial relationships of the components in 3D memory device **100**. Substrate **101** includes two lateral surfaces extending laterally in the x-y plane: a front surface on the front side of the wafer, and a back surface on the backside opposite to the front side of the wafer. The x- and y-directions

are two orthogonal directions in the wafer plane: x-direction is the word line direction, and the y-direction is the bit line direction. The z-axis is perpendicular to both the x- and y-axes. As used herein, whether one component (e.g., a layer or a device) is “on,” “above,” or “below” another component (e.g., a layer or a device) of a semiconductor device (e.g., 3D memory device **100**) is determined relative to the substrate of the semiconductor device (e.g., substrate **101**) in the z-direction (the vertical direction perpendicular to the x-y plane) when the substrate is positioned in the lowest plane of the semiconductor device in the z-direction. The same notion for describing spatial relationships is applied throughout the present disclosure.

(22) In some embodiments, peripheral circuit **108** is configured to control and sense the 3D memory device **100**. Peripheral circuit **108** can be any suitable digital, analog, and/or mixed-signal control and sensing circuits used for facilitating the operation of 3D memory device **100** including, but not limited to, a page buffer, a decoder (e.g., a row decoder and a column decoder), a sense amplifier, a driver (e.g., a word line driver), a charge pump, a current or voltage reference, or any active or passive components of the circuit (e.g., transistors, diodes, resistors, or capacitors). Peripheral circuits **108** can include transistors formed “on” substrate **101**, in which the entirety or part of the transistors are formed in substrate **101** (e.g., below the top surface of substrate **101**) and/or directly on substrate **101**. Isolation regions (e.g., shallow trench isolations (STIs)) and doped regions (e.g., source regions and drain regions of the transistors) can be formed in substrate **101** as well. The transistors are high-speed with advanced logic processes (e.g., technology nodes of 90 nm, 65 nm, 45 nm, 32 nm, 28 nm, 20 nm, 16 nm, 14 nm, 10 nm, 7 nm, 5 nm, 3 nm, etc.), according to some embodiments. It is understood that in some embodiments, peripheral circuit **108** may further include any other circuits compatible with the advanced logic processes including logic circuits, such as processors and programmable logic devices (PLDs), or memory circuits, such as static random-access memory (SRAM).

(23) In some embodiments, first semiconductor structure **102** of 3D memory device **100** further includes an interconnect layer (not shown) above peripheral circuits **108** to transfer electrical signals to and from peripheral circuits **108**. The interconnect layer can include a plurality of interconnects (also referred to herein as “contacts”), including lateral interconnect lines and vertical interconnect access (VIA) contacts. As used herein, the term “interconnects” can broadly include any suitable types of interconnects, such as middle-end-of-line (MEOL) interconnects and back-end-of-line (BEOL) interconnects. The interconnect layer can further include one or more interlayer dielectric (ILD) layers (also known as “intermetal dielectric (IMD) layers”) in which the interconnect lines and VIA contacts can form. That is, the interconnect layer can include interconnect lines and VIA contacts in multiple ILD layers. The interconnect lines and VIA contacts in the interconnect layer can include conductive materials including, but not limited to, tungsten (W), cobalt (Co), copper (Cu), aluminum (Al), silicides, or any combination thereof. The ILD layers in the interconnect layer can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low dielectric constant (low-k) dielectrics, or any combination thereof.

(24) As shown in FIG. 1A, first semiconductor structure **102** of 3D memory device **100** can further include a bonding layer **110** at bonding interface **106** and above the interconnect layer and peripheral circuits **108**. Bonding layer **110** can include a plurality of bonding contacts **111** and dielectrics electrically isolating bonding contacts **111**. Bonding contacts **111** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The remaining area of bonding layer **110** can be formed with dielectrics including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. Bonding contacts **111** and surrounding dielectrics in bonding layer **110** can be used for hybrid bonding.

(25) Similarly, as shown in FIG. 1A, second semiconductor structure **104** of 3D memory device **100** can also include a bonding layer **112** at bonding interface **106** and above bonding layer **110** of

first semiconductor structure **102**. Bonding layer **112** can include a plurality of bonding contacts **113** and dielectrics electrically isolating bonding contacts **113**. Bonding contacts **113** can include conductive materials including, but not limited to, W, Co, Cu, Al, silicides, or any combination thereof. The remaining area of bonding layer **112** can be formed with dielectrics including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. Bonding contacts **113** and surrounding dielectrics in bonding layer **112** can be used for hybrid bonding. Bonding contacts **113** are in contact with bonding contacts **111** at bonding interface **106**, according to some embodiments.

(26) As described below in detail, second semiconductor structure **104** can be bonded on top of first semiconductor structure **102** in a face-to-face manner at bonding interface **106**. In some embodiments, bonding interface **106** is disposed between bonding layers **110** and **112** as a result of hybrid bonding (also known as “metal/dielectric hybrid bonding”), which is a direct bonding technology (e.g., forming bonding between surfaces without using intermediate layers, such as solder or adhesives) and can obtain metal-metal bonding and dielectric-dielectric bonding simultaneously. In some embodiments, bonding interface **106** is the place at which bonding layers **112** and **110** are met and bonded. In practice, bonding interface **106** can be a layer with a certain thickness that includes the top surface of bonding layer **110** of first semiconductor structure **102** and the bottom surface of bonding layer **112** of second semiconductor structure **104**.

(27) In some embodiments, second semiconductor structure **104** of 3D memory device **100** further includes an interconnect layer (not shown) above bonding layer **112** to transfer electrical signals. The interconnect layer can include a plurality of interconnects, such as MEOL interconnects and BEOL interconnects. The interconnect layer can further include one or more ILD layers in which the interconnect lines and VIA contacts can form. The interconnect lines and VIA contacts in the interconnect layer can include conductive materials including, but not limited to W, Co, Cu, Al, silicides, or any combination thereof. The ILD layers in the interconnect layer can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof.

(28) In some embodiments, 3D memory device **100** is a NAND Flash memory device in which memory cells are provided in the form of an array of NAND memory strings. As shown in FIG. **1A**, second semiconductor structure **104** of 3D memory device **100** can include an array of channel structures **124** functioning as the array of NAND memory strings. As shown in FIG. **1A**, each channel structure **124** can extend vertically through a plurality of pairs each including a stack conductive layer **116** and a stack dielectric layer **118**. The interleaved stack conductive layers **116** and stack dielectric layers **118** are part of a memory stack **114**. The number of the pairs of stack conductive layers **116** and stack dielectric layers **118** in memory stack **114** (e.g., 32, 64, 96, 128, 160, 192, 224, 256, or more) determines the number of memory cells in 3D memory device **100**. It is understood that in some embodiments, memory stack **114** may have a multi-deck architecture (not shown), which includes a plurality of memory decks stacked over one another. The numbers of the pairs of stack conductive layers **116** and stack dielectric layers **118** in each memory deck can be the same or different.

(29) Memory stack **114** can include a plurality of interleaved stack conductive layers **116** and stack dielectric layers **118**. Stack conductive layers **116** and stack dielectric layers **118** in memory stack **114** can alternate in the vertical direction. In other words, except the ones at the top or bottom of memory stack **114**, each stack conductive layer **116** can be adjoined by two stack dielectric layers **118** on both sides, and each stack dielectric layer **118** can be adjoined by two stack conductive layers **116** on both sides. Stack conductive layers **116** can include conductive materials including, but not limited to, W, Co, Cu, Al, polysilicon, doped silicon, silicides, or any combination thereof. Each stack conductive layer **116** can include a gate electrode (gate line) surrounded by an adhesive layer and a gate dielectric layer. The gate electrode of stack conductive layer **116** can extend laterally as a word line, ending at one or more staircase structures of memory stack **114**. Stack

dielectric layers **118** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, or any combination thereof.

(30) As shown in FIG. 1A, second semiconductor structure **104** of 3D memory device **100** can also include a first semiconductor layer **120** above memory stack **114** and a second semiconductor layer **122** above and in contact with first semiconductor layer **120**. In some embodiments, each of first and second semiconductor layers **120** and **122** is an N-type doped semiconductor layer, e.g., a silicon layer doped with N-type dopant(s), such as phosphorus (P) or arsenic (As). In those cases, first and second semiconductor layers **120** and **122** may be viewed collectively as an N-type doped semiconductor layer **120/122** above memory stack **114**. In some embodiments, each of first and second semiconductor layers **120** and **122** includes an N-well. That is, each of first and second semiconductor layers **120** and **122** can be a region in a P-type substrate that is doped with N-type dopant(s), such as P or As. It is understood that the doping concentrations in first and second semiconductor layers **120** and **122** may be the same or different. First semiconductor layer **120** includes polysilicon, for example, N-type doped polysilicon, according to some embodiments. As described below in detail, first semiconductor layer **120** can be formed above a P-type silicon substrate by thin film deposition and/or epitaxial growth. In contrast, second semiconductor layer **122** includes single crystalline silicon, for example, N-type doped single crystalline silicon, according to some embodiments. As described below in detail, second semiconductor layer **122** can be formed by implanting N-type dopant(s) into a P-type silicon substrate having single crystalline silicon.

(31) In some embodiments, each channel structure **124** includes a channel hole filled with a semiconductor layer (e.g., as a semiconductor channel **128**) and a composite dielectric layer (e.g., as a memory film **126**). In some embodiments, semiconductor channel **128** includes silicon, such as amorphous silicon, polysilicon, or single crystalline silicon. For example, semiconductor channel **128** may include polysilicon. In some embodiments, memory film **126** is a composite layer including a tunneling layer, a storage layer (also known as a “charge trap layer”), and a blocking layer. The remaining space of channel structure **124** can be partially or fully filled with a capping layer including dielectric materials, such as silicon oxide, and/or an air gap. Channel structure **124** can have a cylinder shape (e.g., a pillar shape). The capping layer, semiconductor channel **128**, the tunneling layer, storage layer, and blocking layer of memory film **126** are arranged radially from the center toward the outer surface of the pillar in this order, according to some embodiments. The tunneling layer can include silicon oxide, silicon oxynitride, or any combination thereof. The storage layer can include silicon nitride, silicon oxynitride, silicon, or any combination thereof. The blocking layer can include silicon oxide, silicon oxynitride, high-k dielectrics, or any combination thereof. In one example, memory film **126** can include a composite layer of silicon oxide/silicon oxynitride/silicon oxide (ONO).

(32) In some embodiments, channel structure **124** further includes a channel plug **129** in the bottom portion (e.g., at the lower end) of channel structure **124**. As used herein, the “upper end” of a component (e.g., channel structure **124**) is the end farther away from substrate **101** in the z-direction, and the “lower end” of the component (e.g., channel structure **124**) is the end closer to substrate **101** in the z-direction when substrate **101** is positioned in the lowest plane of 3D memory device **100**. Channel plug **129** can include semiconductor materials (e.g., polysilicon). In some embodiments, channel plug **129** functions as the drain of the NAND memory string.

(33) As shown in FIG. 1A, each channel structure **124** can extend vertically through interleaved stack conductive layers **116** and stack dielectric layers **118** of memory stack **114** and first semiconductor layer **120**, e.g., an N-type doped polysilicon layer. In some embodiments, first semiconductor layer **120** surrounds part of channel structure **124** and is in contact with semiconductor channel **128** including polysilicon. That is, memory film **126** is disconnected at part of channel structure **124** that abuts first semiconductor layer **120**, exposing semiconductor channel **128** to be in contact with the surrounding first semiconductor layer **120**, according to some

embodiments. As a result, first semiconductor layer **120** surrounding and in contact with semiconductor channel **128** can work as a “sidewall semiconductor plug” of channel structure **124**. (34) In some embodiments, each channel structure **124** can extend vertically further into second semiconductor layer **122**, e.g., an N-type doped single crystalline silicon layer. That is, each channel structure **124** extends vertically through memory stack **114** into the N-type doped semiconductor layer (including first and second semiconductor layers **120** and **122**), according to some embodiments. As shown in FIG. **1A**, the top portion (e.g., the upper end) of channel structures **124** is in second semiconductor layer **122**, according to some embodiments. In some embodiments, each of first and second semiconductor layers **120** and **122** is an N-type doped semiconductor layer, e.g., an N-well, to enable GIDL-assisted body biasing for erase operations. The GIDL around the source select gate of the NAND memory string can generate hole current into the NAND memory string to raise the body potential for erase operations.

(35) It is understood that first and second semiconductor layers **120** and **122** illustrate one example of a semiconductor layer that can be used for 3D memory device **100**. In a more general case, second semiconductor structure **104** of 3D memory device **100** may include a semiconductor layer having one or more doped silicon layers, such as doped polysilicon layer(s) and/or doped single crystalline silicon layer(s). Moreover, the number of the doped silicon layers in the semiconductor layer and the doping type of each silicon layer in the semiconductor layer are not limited by the example above with respect to FIG. **1A** and may vary in other examples. It is also understood that the relative position of channel structure **124** with respect to the semiconductor layer is not limited by the example above with respect to FIG. **1A** and may vary in other examples. In a more general case, each channel structure **124** may extend vertically through memory stack **114** into the semiconductor layer regardless of the relative position of the end of channel structure **124** with respect to each doped silicon layer in the semiconductor layer. It is further understood that the erase operation performed by 3D memory device **100** is not limited to GIDL erase in the example above with respect to FIG. **1A** and may be P-well bulk erase operation or any other suitable erase operations based on different configurations of the semiconductor layer, for example, the doping type of each doped silicon layer in the semiconductor layer.

(36) Compared with known 3D memory devices with polysilicon semiconductor channels that have defects (e.g., dangling bonds) between grain boundaries and at the interfaces with the tunneling layers, 3D memory device **100** can include hydrogen-rich polysilicon semiconductor channel **128** in each channel structure **124**, which have fewer defects (e.g., dangling bonds). In some embodiments, semiconductor channel **128** is doped with at least one of hydrogen or an isotope of hydrogen, such as protium, deuterium, tritium, hydrogen-4, hydrogen-5, etc. The hydrogen or isotope thereof can form a stable H—Si covalent bond with the dangling bonds between the grain boundaries and at the interface with the tunneling layer to reduce the number of dangling bonds, thereby improving the carrier mobility and adjusting the stress of semiconductor channel **128**.

(37) As shown in FIG. **1A**, second semiconductor structure **104** of 3D memory device **100** can further include insulating structures **130** each extending vertically through interleaved stack conductive layers **116** and stack dielectric layers **118** of memory stack **114**. Different from channel structure **124** that extends further through first semiconductor layer **120**, insulating structures **130** stops at first semiconductor layer **120**, i.e., does not extend vertically into the N-type doped semiconductor layer, according to some embodiments. That is, the top surface of insulating structure **130** can be flush with the bottom surface of first semiconductor layer **120**. Each insulating structure **130** can also extend laterally to separate channel structures **124** into a plurality of blocks. That is, memory stack **114** can be divided into a plurality of memory blocks by insulating structures **130**, such that the array of channel structures **124** can be separated into each memory block. It is understood that the relative position of insulating structure **130** with respect to the semiconductor layer is not limited by the example above with respect to FIG. **1A** and may vary in other examples. For example, insulating structure **130** may extend further into the semiconductor layer.

(38) In some embodiments, insulating structure **130** does not include any contact therein (i.e., not functioning as the source contact) and thus, does not introduce parasitic capacitance and leakage current with conductive layers **116** (including word lines), according to some embodiments. In some embodiments, each insulating structure **130** includes an opening (e.g., a slit) filled with one or more dielectric materials, including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, or any combination thereof. As shown in FIG. 1A, insulating structure **130** can include a dielectric layer **160** doped with at least one of hydrogen or an isotope of hydrogen, such as protium, deuterium, tritium, hydrogen-4, hydrogen-5, etc. That is, 3D memory device **100** can include hydrogen-rich insulating structures **130** as the source for providing hydrogen or an isotope thereof to semiconductor channels **128** as described below in detail with respect to the fabrication process of 3D memory device **100**.

(39) In some embodiments, the atomic percent of the at least one of hydrogen or an isotope of hydrogen in dielectric layer **160** is not greater than about 50%, such as not greater than 50%. In some embodiments, the atomic percent of the at least one of hydrogen or an isotope of hydrogen in dielectric layer **160** is between about 0.1% and about 50%, such as between 0.1% and 50% (e.g., 0.1%, 0.2%, 0.3%, 0.4%, 0.5%, 0.6%, 0.7%, 0.8%, 0.9%, 1%, 5%, 10%, 15%, 20%, 25%, 30%, 35%, 40%, 45%, 50%, any range bounded by the lower end by any of these values, or in any range defined by any two of these values). In one example, the atomic percent of the at least one of hydrogen or an isotope of hydrogen in dielectric layer **160** may be between about 1% and about 10%, such as between 1% and 10% (e.g., 1%, 2%, 3%, 4%, 5%, 6%, 7%, 8%, 9%, 10%, any range bounded by the lower end by any of these values, or in any range defined by any two of these values). Dielectric layer **160** can include any dielectric materials that can absorb hydrogen or the isotope thereof, such as tetraethyl orthosilicate (TEOS) or silicon nitride. In some embodiments, dielectric layer **160** includes silicon nitride, which has a strong hydrogen absorption capability.

(40) It is understood that the structure of hydrogen-rich insulating structures **130** is not limited to the example shown in FIG. 1A. FIGS. 2A-2H illustrate side views of cross-sections of various exemplary hydrogen-rich insulating structures **130** in 3D memory device **100** in FIG. 1A, according to various embodiments of the present disclosure. As shown in FIG. 2A, in some embodiments, insulating structure **130** includes a first silicon oxide layer **202**, dielectric layer **160**, and a second silicon oxide layer **204**. First silicon oxide layer **202** can be disposed laterally between dielectric layer **160** and stack conductive layers **116** of memory stack **114**. In some embodiments, first silicon oxide layer **202** is disposed along the sidewalls and bottom surface of insulating structure **130** and is in contact with stack conductive layers **116** of memory stack **114**. As described above, dielectric layer **160**, such as a silicon nitride layer, can be doped with at least one of hydrogen or an isotope of hydrogen. In some embodiments, dielectric layer **160** is disposed over first silicon oxide layer **202**, e.g., along the sidewalls and bottom surface of insulating structure **130**. Second silicon oxide layer **204** can fill the remaining space of insulating structure **130** either with or without an air gap therein. In some embodiments, dielectric layer **160** is disposed laterally between first and second silicon oxide layers **202** and **204**. In some embodiments, one end of insulating structure **130** is flush with the respective end of dielectric layer **160**. For example, the lower end of insulating structure **130** may be flush with the lower end of dielectric layer **160**. As shown in FIG. 2A, the lower ends of first and second silicon oxide layers **202** and **204** and dielectric layer **160** are flush with one another, i.e., coplanar, at the lower end of insulating structure **130**.

(41) In some embodiments, one end of insulating structure **130** is not flush with the respective end of dielectric layer **160**, but instead, includes a silicon oxide cap **208** at the end thereof. As shown in FIG. 2B, different from the example shown in FIG. 2A, insulating structure **130** can include silicon oxide cap **208** at the lower end thereof. In some embodiments, silicon oxide cap **208** is disposed below the lower end of dielectric layer **160** and is connected to first and second silicon oxide layers **202** and **204**. As a result, silicon oxide cap **208** can prevent hydrogen or its isotope from diffusing

away from the lower end of insulating structure **130**, which may reduce the amount of hydrogen or its isotope that can be diffused laterally through the sidewalls of insulating structure **130** to channel structures **124**.

(42) In some embodiments, insulating structure **130** does not include second silicon oxide layer **204**. As shown in FIGS. 2C and 2D, dielectric layer **160** can fill the remaining space of insulating structure **130** either with or without an air gap therein, e.g., replacing second silicon oxide layer **204** in FIGS. 2A and 2B. Similar to FIGS. 2A and 2B, insulating structure **130** may or may not include silicon oxide cap **208** at the lower end thereof. Compared with the examples in FIGS. 2A and 2B, dielectric layer **160** in FIGS. 2C and 2D can have a larger volume of dielectric materials, such as silicon nitride, to absorb more hydrogen or its isotope. Moreover, by changing the material filling the remaining space of insulating structure **130**, e.g., from silicon oxide to silicon nitride, the stress of insulating structure **130** may be adjusted as well, which can further adjust the wafer stress as needed.

(43) In some embodiments, insulating structure **130** does not include first and second silicon oxide layers **202** and **204**. As shown in FIGS. 2E and 2F, dielectric layer **160** can fill the space of insulating structure **130** either with or without an air gap therein, e.g., replacing both first and second silicon oxide layers **202** and **204** in FIGS. 2A and 2B. That is, dielectric layer **160** can be in contact with stack conductive layers **116** of memory stack **114**. Similar to FIGS. 2A and 2B, insulating structure **130** may or may not include silicon oxide cap **208** at the lower end thereof. Compared with the examples in FIGS. 2A and 2B, dielectric layer **160** in FIGS. 2E and 2F can have a larger volume of dielectric materials, such as silicon nitride, to absorb more hydrogen or its isotope. Moreover, by changing the material filling the space of insulating structure **130**, e.g., from silicon oxide to silicon nitride, the stress of insulating structure **130** may be adjusted as well, which can further adjust the wafer stress as needed.

(44) In some embodiments, insulating structure **130** does not include second silicon oxide layer **204**, but instead includes a polysilicon layer **210**, replacing second silicon oxide layer **204**. As shown in FIGS. 2G and 2H, polysilicon layer **210** can fill the remaining space of insulating structure **130** either with or without an air gap therein, e.g., replacing second silicon oxide layer **204** in FIGS. 2A and 2B. In some embodiments, dielectric layer **160** is disposed laterally between first silicon oxide layer **202** and polysilicon layer **210**. Similar to FIGS. 2A and 2B, insulating structure **130** may or may not include silicon oxide cap **208** at the lower end thereof. By changing the material filling the remaining space of insulating structure **130**, e.g., from silicon oxide to polysilicon, the stress of insulating structure **130** may be adjusted as well, which can further adjust the wafer stress as needed.

(45) It is understood that the structure of insulating structure **130** may further vary in other examples, for example, by changing the materials filling the remaining space of insulating structure **130**, the number of layers in insulating structure **130**, and so on, in order to adjust the amount of hydrogen or its isotope that can be absorbed by dielectric layer **160** and/or adjust the stress of insulating structure **130** as needed. It is also understood that insulating structure **130** may include additional layers, such as a high-k dielectric layer **206** shown in FIGS. 2A-2H. High-k dielectric layer **206** may extend from the gate dielectric layers of stack conductive layers **116**.

(46) Referring back to FIG. 1A, 3D memory device **100** can include a backside source contact **132** above memory stack **114** and in contact with second semiconductor layer **122**. Source contact **132** and memory stack **114** (and insulating structure **130** therethrough) can be disposed on opposite sides of the semiconductor layer (e.g., a thinned substrate) and thus, viewed as a “backside” source contact. In some embodiments, source contact **132** extends further into second semiconductor layer **122** and is electrically connected to first semiconductor layer **120** and semiconductor channel **128** of channel structure **124** through second semiconductor layer **122**. It is understood that the depth that source contact **132** extends into second semiconductor layer **122** may vary in different examples. In some embodiments in which second semiconductor layer **122** includes an N-well,

source contact **132** is also referred to herein as an “N-well pick up.” Source contacts **132** can include any suitable types of contacts. In some embodiments, source contacts **132** include a VIA contact. In some embodiments, source contacts **132** include a wall-shaped contact extending laterally. Source contact **132** can include one or more conductive layers, such as a metal layer (e.g., W, Co, Cu, or Al) or a silicide layer surrounded by an adhesive layer (e.g., titanium nitride (TiN)). (47) As shown in FIG. 1A, 3D memory device **100** can further include a BEOL interconnect layer **133** above and in contact with source contact **132** for pad-out, e.g., transferring electrical signals between 3D memory device **100** and external circuits. In some embodiments, interconnect layer **133** includes one or more ILD layers **134** on second semiconductor layer **122** and a redistribution layer **136** on ILD layers **134**. The upper end of source contact **132** is flush with the top surface of ILD layers **134** and the bottom surface of redistribution layer **136**, and source contact **132** extends vertically through ILD layers **134** into second semiconductor layer **122**, according to some embodiments. ILD layers **134** in interconnect layer **133** can include dielectric materials including, but not limited to, silicon oxide, silicon nitride, silicon oxynitride, low-k dielectrics, or any combination thereof. Redistribution layer **136** in interconnect layer **133** can include conductive materials including, but not limited to W, Co, Cu, Al, silicides, or any combination thereof. In some embodiments, interconnect layer **133** further includes a passivation layer **138** as the outmost layer for passivation and protection of 3D memory device **100**. Part of redistribution layer **136** can be exposed from passivation layer **138** as contact pads **140**. That is, interconnect layer **133** of 3D memory device **100** can also include contact pads **140** for wire bonding and/or bonding with an interposer.

(48) In some embodiments, second semiconductor structure **104** of 3D memory device **100** further includes contacts **142** and **144** through second semiconductor layer **122**. As second semiconductor layer **122** can be a thinned substrate, for example, an N-well of a P-type silicon substrate, contacts **142** and **144** are through silicon contacts (TSCs), according to some embodiments. In some embodiments, contact **142** extends through second semiconductor layer **122** and ILD layers **134** to be in contact with redistribution layer **136**, such that first semiconductor layer **120** is electrically connected to contact **142** through second semiconductor layer **122**, source contact **132**, and redistribution layer **136** of interconnect layer **133**. In some embodiments, contact **144** extends through second semiconductor layer **122** and ILD layers **134** to be in contact with contact pad **140**. Contacts **142** and **144** each can include one or more conductive layers, such as a metal layer (e.g., W, Co, Cu, or Al) or a silicide layer surrounded by an adhesive layer (e.g., TiN). In some embodiments, at least contact **144** further includes a spacer (e.g., a dielectric layer) to electrically insulate contact **144** from second semiconductor layer **122**.

(49) In some embodiments, 3D memory device **100** further includes peripheral contacts **146** and **148** each extending vertically to second semiconductor layer **122** (e.g., an N-well of a P-type silicon substrate) outside of memory stack **114**. Each peripheral contact **146** or **148** can have a depth greater than the depth of memory stack **114** to extend vertically from bonding layer **112** to second semiconductor layer **122** in a peripheral region that is outside of memory stack **114**. In some embodiments, peripheral contact **146** is below and in contact with contact **142**, such that first semiconductor layer **120** is electrically connected to peripheral circuit **108** in first semiconductor structure **102** through at least second semiconductor layer **122**, source contact **132**, interconnect layer **133**, contact **142**, and peripheral contact **146**. In some embodiments, peripheral contact **148** is below and in contact with contact **144**, such that peripheral circuit **108** in first semiconductor structure **102** is electrically connected to contact pad **140** for pad-out through at least contact **144** and peripheral contact **148**. Peripheral contacts **146** and **148** each can include one or more conductive layers, such as a metal layer (e.g., W, Co, Cu, or Al) or a silicide layer surrounded by an adhesive layer (e.g., TiN).

(50) As shown in FIG. 1A, 3D memory device **100** also includes a variety of local contacts (also known as “C1”) as part of the interconnect structure, which are in contact with a structure in

memory stack **114** directly. In some embodiments, the local contacts include channel local contacts **150** each below and in contact with the lower end of a respective channel structure **124**. Each channel local contact **150** can be electrically connected to a bit line contact (not shown) for bit line fan-out. In some embodiments, the local contacts further include word line local contacts **152** each below and in contact with a respective conductive layer **116** (including a word line) at the staircase structure of memory stack **114** for word line fan-out. Local contacts, such as channel local contacts **150** and word line local contacts **152**, can be electrically connected to peripheral circuits **108** of first semiconductor structure **102** through at least bonding layers **112** and **110**. Local contacts, such as channel local contacts **150** and word line local contacts **152**, each can include one or more conductive layers, such as a metal layer (e.g., W, Co, Cu, or Al) or a silicide layer surrounded by an adhesive layer (e.g., TiN).

(51) FIG. **1B** illustrates a side view of a cross-section of another exemplary 3D memory device **103** with hydrogen-rich semiconductor channels, according to some embodiments of the present disclosure. 3D memory device **103** is similar to 3D memory device **100** except that backside source contact **132** in 3D memory device **100** is replaced by a front side source contact **162** in 3D memory device **103**, according to some embodiments. As shown in FIG. **1B**, source contact **162** can be in contact with the semiconductor layer (e.g., second semiconductor layer **122**), and source contact **162** and insulating structure **130** can be disposed on the same side, e.g., the front side, of the semiconductor layer. It is understood that the details of other same structures in both 3D memory devices **103** and **100** are not repeated for ease of description

(52) FIGS. **3A-3M** illustrate a fabrication process for forming an exemplary 3D memory device with hydrogen-rich semiconductor channels, according to some embodiments of the present disclosure. FIG. **4** illustrates a flowchart of a method for forming an exemplary 3D memory device with hydrogen-rich semiconductor channels, according to some embodiments of the present disclosure. Examples of the 3D memory device depicted in FIGS. **3A-3M** and **4** include 3D memory devices **100** and **103** depicted in FIGS. **1A** and **1B**. FIGS. **3A-3M** and **4** will be described together. It is understood that the operations shown in method **400** are not exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. **4**.

(53) Referring to FIG. **4**, method **400** starts at operation **402**, in which a peripheral circuit is formed on a first substrate. The first substrate can be a silicon substrate. As illustrated in FIG. **3I**, a plurality of transistors are formed on a silicon substrate **350** using a plurality of processes including, but not limited to, photolithography, etching, thin film deposition, thermal growth, implantation, chemical mechanical polishing (CMP), and any other suitable processes. In some embodiments, doped regions (not shown) are formed in silicon substrate **350** by ion implantation and/or thermal diffusion, which function, for example, as source regions and/or drain regions of the transistors. In some embodiments, isolation regions (e.g., STIs) are also formed in silicon substrate **350** by wet etching and/or dry etching and thin film deposition. The transistors can form peripheral circuits **352** on silicon substrate **350**.

(54) As illustrated in FIG. **3I**, a bonding layer **348** is formed above peripheral circuits **352**. Bonding layer **348** includes bonding contacts electrically connected to peripheral circuits **352**. To form bonding layer **348**, an ILD layer is deposited using one or more thin film deposition processes, such as chemical vapor deposition (CVD), physical vapor deposition (PVD), atomic layer deposition (ALD), or any combination thereof, and the bonding contacts are formed through the ILD layer using wet etching and/or dry etching, e.g., RIE, followed by one or more thin film deposition processes, such as ALD, CVD, PVD, any other suitable processes, or any combination thereof.

(55) Method **400** proceeds to operation **404**, as illustrated in FIG. **4**, in which a first semiconductor layer on a second substrate, a sacrificial layer on the first semiconductor layer, and a dielectric

stack on the sacrificial layer are sequentially formed. The dielectric stack can include interleaved stack sacrificial layers and stack dielectric layers. In some embodiments, to subsequently form the sacrificial layer and the dielectric stack, polysilicon is deposited on the second semiconductor layer to form the sacrificial layer, and stack dielectric layers and stack sacrificial layers are alternatingly deposited on the sacrificial layer to form the dielectric stack.

(56) As illustrated in FIG. 3A, an N-type doped semiconductor layer **304** is formed on a silicon substrate **302**. N-type doped semiconductor layer **304** can include an N-well in a P-type silicon substrate **302** and include single crystalline silicon. N-type doped semiconductor layer **304** can be formed by doping N-type dopant(s), such as P or As, into P-type silicon substrate **302** using ion implantation and/or thermal diffusion.

(57) As illustrated in FIG. 3A, a sacrificial layer **306** is formed on N-type doped semiconductor layer **304**. Sacrificial layer **306** can be formed by depositing polysilicon or any other suitable sacrificial material (e.g., silicon nitride, carbon) that can be later selectively removed using one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. In some embodiments, a pad oxide layer **305** is formed between sacrificial layer **306** and N-type doped semiconductor layer **304** by depositing dielectric materials, such as silicon oxide, or thermal oxidation, on silicon substrate **302** prior to the formation of N-type doped semiconductor layer **304**.

(58) As illustrated in FIG. 3A, a dielectric stack **308** including a plurality pairs of a first dielectric layer (referred to herein as “stack sacrificial layer” **312**) and a second dielectric layer (referred to herein as “stack dielectric layers” **310**, together referred to herein as “dielectric layer pairs”) is formed on sacrificial layer **306**. Dielectric stack **308** includes interleaved stack sacrificial layers **312** and stack dielectric layers **310**, according to some embodiments. Stack dielectric layers **310** and stack sacrificial layers **312** can be alternatively deposited on sacrificial layer **306** above silicon substrate **302** to form dielectric stack **308**. In some embodiments, each stack dielectric layer **310** includes a layer of silicon oxide, and each stack sacrificial layer **312** includes a layer of silicon nitride. Dielectric stack **308** can be formed by one or more thin film deposition processes including, but not limited to, CVD, PVD, ALD, or any combination thereof. As illustrated in FIG. 3A, a staircase structure can be formed on the edge of dielectric stack **308**. The staircase structure can be formed by performing a plurality of so-called “trim-etch” cycles to the dielectric layer pairs of dielectric stack **308** toward silicon substrate **302**. Due to the repeated trim-etch cycles applied to the dielectric layer pairs of dielectric stack **308**, dielectric stack **308** can have one or more tilted edges and a top dielectric layer pair shorter than the bottom one, as shown in FIG. 3A.

(59) Method **400** proceeds to operation **406**, as illustrated in FIG. 4, in which a channel structure extending vertically through the dielectric stack and the sacrificial layer into the first semiconductor layer is formed. In some embodiments, to form the channel structure, a channel hole extending vertically through the dielectric stack and the sacrificial layer into the first semiconductor layer is formed, a memory film and a semiconductor channel are sequentially formed over a sidewall of the channel hole, and a channel plug is formed above and in contact with the semiconductor channel. The semiconductor channel can include polysilicon.

(60) As illustrated in FIG. 3A, a channel hole is an opening extending vertically through dielectric stack **308** and sacrificial layer **306** into N-type doped semiconductor layer **304**. In some embodiments, a plurality of openings are formed, such that each opening becomes the location for growing an individual channel structure **314** in the later process. In some embodiments, fabrication processes for forming the channel hole of channel structure **314** include wet etching and/or dry etching, such as deep-ion reactive etching (DRIE). In some embodiments, the channel hole of channel structure **314** extends further through the top portion of N-type doped semiconductor layer **304**. The etching process through dielectric stack **308** and sacrificial layer **306** may continue to etch part of N-type doped semiconductor layer **304**. In some embodiments, a separate etching process is used to etch part of N-type doped semiconductor layer **304** after etching through dielectric stack

308 and sacrificial layer **306**.

(61) As illustrated in FIG. 3A, a memory film **316** (including a blocking layer, a storage layer, and a tunneling layer) and a semiconductor channel **318** are sequentially formed in this order along sidewalls and the bottom surface of the channel hole. In some embodiments, memory film **316** is first deposited along the sidewalls and bottom surface of the channel hole, and semiconductor channel **318** is then deposited over memory film **316**. The blocking layer, storage layer, and tunneling layer can be subsequently deposited in this order using one or more thin film deposition processes, such as ALD, CVD, PVD, any other suitable processes, or any combination thereof, to form memory film **316**. Semiconductor channel **318** can then be formed by depositing a semiconductor material, such as polysilicon, over the tunneling layer of memory film **316** using one or more thin film deposition processes, such as ALD, CVD, PVD, any other suitable processes, or any combination thereof. In some embodiments, a first silicon oxide layer, a silicon nitride layer, a second silicon oxide layer, and a polysilicon layer (a “SONO” structure) are subsequently deposited to form memory film **316** and semiconductor channel **318**.

(62) As illustrated in FIG. 3A, a capping layer is formed in the channel hole and over semiconductor channel **318** to completely or partially fill the channel hole (e.g., without or with an air gap). The capping layer can be formed by depositing a dielectric material, such as silicon oxide, using one or more thin film deposition processes, such as ALD, CVD, PVD, any other suitable processes, or any combination thereof. A channel plug can then be formed in the top portion of the channel hole. In some embodiments, parts of memory film **316**, semiconductor channel **318**, and the capping layer that are on the top surface of dielectric stack **308** are removed and planarized by CMP, wet etching, and/or dry etching. A recess then can be formed in the top portion of the channel hole by wet etching and/or drying etching parts of semiconductor channel **318** and the capping layer in the top portion of the channel hole. The channel plug can then be formed by depositing semiconductor materials, such as polysilicon, into the recess by one or more thin film deposition processes, such as CVD, PVD, ALD, or any combination thereof. Channel structure **314** is thereby formed through dielectric stack **308** and sacrificial layer **306** into N-type doped semiconductor layer **304**.

(63) Method **400** proceeds to operation **408**, as illustrated in FIG. 4, in which an opening extending vertically through the dielectric stack is formed, for example, to expose part of the sacrificial layer. As illustrated in FIG. 3A, a slit **320** is an opening that extends vertically through dielectric stack **308** and exposes part of sacrificial layer **306**. In some embodiments, fabrication processes for forming slit **320** include wet etching and/or dry etching, such as DRIE. In some embodiments, slit **320** extends further into the top portion of sacrificial layer **306**. The etching process through dielectric stack **308** may not stop at the top surface of sacrificial layer **306** and may continue to etch part of sacrificial layer **306**.

(64) Method **400** proceeds to operation **410**, as illustrated in FIG. 4, in which the sacrificial layer is replaced with a second semiconductor layer through the opening. As illustrated in FIG. 3B, sacrificial layer **306** (shown in FIG. 3A) is removed by wet etching and/or dry etching to form a cavity **322**. In some embodiments, sacrificial layer **306** is wet etched by applying a wet etchant through slit **320**, which can be stopped by pad oxide layer **305** between sacrificial layer **306** and N-type doped semiconductor layer **304**. That is, the removal of sacrificial layer **306** does not affect N-type doped semiconductor layer **304**, according to some embodiments. In some embodiments, prior to the removal of sacrificial layer **306**, a spacer **324** is formed along the sidewall of slit **320**. Spacer **324** can be formed by depositing dielectric materials, such as silicon nitride, silicon oxide, and silicon nitride, into slit **320** using one or more thin film deposition processes, such as CVD, PVD, ALD, or any combination thereof.

(65) As illustrated in FIG. 3C, part of memory film **316** of channel structure **314** exposed in cavity **322** is removed to expose part of semiconductor channel **318** of channel structure **314** abutting cavity **322**. In some embodiments, parts of the blocking layer (e.g., including silicon oxide),

storage layer (e.g., including silicon nitride), and tunneling layer (e.g., including silicon oxide) are etched by applying etchants through slit **320** and cavity **322**, for example, phosphoric acid for etching silicon nitride and hydrofluoric acid for etching silicon oxide. The etching can be stopped by semiconductor channel **318** of channel structure **314**. Spacer **324** including dielectric materials (shown in FIG. **3B**) can also protect dielectric stack **308** from the etching of memory film **316** and can be removed by the etchants in the same step as removing part of memory film **316**. Similarly, pad oxide layer **305** (shown in FIG. **3B**) on N-type doped semiconductor layer **304** can be removed as well by the same step as removing part of memory film **316**.

(66) As illustrated in FIG. **3D**, an N-type doped semiconductor layer **326** is formed above and in contact with N-type doped semiconductor layer **304**. In some embodiments, N-type doped semiconductor layer **326** is formed by depositing polysilicon into cavity **322** (shown in FIG. **3C**) through slit **320** using one or more thin film deposition processes, such as CVD, PVD, ALD, or any combination thereof. In some embodiments, in-situ doping of N-type dopants, such as P or As, is performed when depositing polysilicon to form an N-type doped polysilicon layer as N-type doped semiconductor layer **326**. N-type doped semiconductor layer **326** can fill cavity **322** to be in contact with the exposed part of semiconductor channel **318** of channel structure **314**.

(67) Method **400** proceeds to operation **412**, as illustrated in FIG. **4**, in which the dielectric stack is replaced with a memory stack, for example, using the so-called “gate replacement” process, such that the channel structure extends vertically through the memory stack and the first semiconductor layer into the second semiconductor layer. In some embodiments, to replace the dielectric stack with the memory stack, the stack sacrificial layers are replaced with stack conductive layers through the opening. In some embodiments, the memory stack includes interleaved stack conductive layers and stack dielectric layers. The channel structure extending vertically through the memory stack into the semiconductor layer on the second substrate is thereby formed, according to some embodiments.

(68) As illustrated in FIG. **3E**, stack sacrificial layers **312** (shown in FIG. **3A**) are replaced with stack conductive layers **328**, and a memory stack **330** including interleaved stack conductive layers **328** and stack dielectric layers **310** is thereby formed, replacing dielectric stack **308** (shown in FIG. **3A**). In some embodiments, lateral recesses (not shown) are first formed by removing stack sacrificial layers **312** through slit **320**. In some embodiments, stack sacrificial layers **312** are removed by applying etchants through slit **320**, creating the lateral recesses interleaved between stack dielectric layers **310**. The etchants can include any suitable etchants that etch stack sacrificial layers **312** selective to stack dielectric layers **310**. As illustrated in FIG. **3E**, stack conductive layers **328** (including gate electrodes and adhesive layers) are deposited into the lateral recesses through slit **320**. In some embodiments, a gate dielectric layer **332** is deposited into the lateral recesses prior to stack conductive layers **328**, such that stack conductive layers **328** are deposited on the gate dielectric layer. Stack conductive layers **328**, such as metal layers, can be deposited using one or more thin film deposition processes, such as ALD, CVD, PVD, any other suitable processes, or any combination thereof. In some embodiments, gate dielectric layer **332**, such as a high-k dielectric layer, is formed along the sidewall and at the bottom of slit **320** as well. Channel structure **314** extending vertically through memory stack **330** into a semiconductor layer (including N-type doped semiconductor layers **304** and **306**) on silicon substrate **302** is thereby formed, according to some embodiments.

(69) Method **400** proceeds to operation **414**, as illustrated in FIG. **4**, in which an insulating structure extending vertically through the memory stack is formed in the opening. The insulating structure can include a dielectric layer doped with at least one of hydrogen or an isotope of hydrogen. In some embodiments, to form the insulating structure, a first layer of silicon oxide, the dielectric layer, and a second layer of silicon oxide are sequentially formed in the opening. In some embodiments, to form the insulating structure, a layer of silicon oxide, the dielectric layer, and a layer of polysilicon are sequentially formed in the opening. In some embodiments, to form the

insulating structure, a layer of silicon oxide and the dielectric layer are sequentially formed in the opening. In some embodiments, to form the dielectric layer, a layer of silicon nitride is deposited, and the layer of silicon nitride is in situ doped with the least one of hydrogen or an isotope of hydrogen. In some embodiments, to form the dielectric layer, a layer of silicon nitride is deposited, and the layer of silicon nitride is annealed in the presence of the least one of hydrogen or an isotope of hydrogen after the deposition. In some embodiments, to form the insulating structure, a silicon oxide cap is formed at an upper end of the insulating structure. In some embodiments, to form the silicon oxide cap, a top portion of at least the dielectric layer is etched back to form a recess, and a layer of silicon oxide is deposited to fill the recess.

(70) As illustrated in FIG. 3F, an insulating structure **336** extending vertically through memory stack **330** is formed, stopping on the top surface of N-type doped semiconductor layer **326**.

Insulating structure **336** can include a dielectric layer **334** doped with least one of hydrogen or an isotope of hydrogen, such as protium, deuterium, tritium, hydrogen-4, hydrogen-5, etc. Referring to FIGS. 2A and 2B, insulating structure **130** (e.g., insulating structure **336** in FIG. 3F) can be formed by sequentially depositing a first silicon oxide layer **202**, dielectric layer **160** (e.g., dielectric layer **334** in FIG. 3F), and a second silicon oxide layer **204** into slit **320** to fully or partially fill slit **320** (with or without an air gap) using one or more thin film deposition processes, such as ALD, CVD, PVD, any other suitable processes, or any combination thereof. Referring to FIGS. 2C and 2D, insulating structure **130** (e.g., insulating structure **336** in FIG. 3F) can be formed by sequentially depositing a first silicon oxide layer **202** and dielectric layer **160** (e.g., dielectric layer **334** in FIG. 3F) into slit **320** to fully or partially fill slit **320** (with or without an air gap) using one or more thin film deposition processes, such as ALD, CVD, PVD, any other suitable processes, or any combination thereof. Referring to FIGS. 2E and 2F, insulating structure **130** (e.g., insulating structure **336** in FIG. 3F) can be formed by depositing dielectric layer **160** (e.g., dielectric layer **334** in FIG. 3F) into slit **320** to fully or partially fill slit **320** (with or without an air gap) using one or more thin film deposition processes, such as ALD, CVD, PVD, any other suitable processes, or any combination thereof. Referring to FIGS. 2G and 2H, insulating structure **130** (e.g., insulating structure **336** in FIG. 3F) can be formed by sequentially depositing a first silicon oxide layer **202**, dielectric layer **160** (e.g., dielectric layer **334** in FIG. 3F), and a polysilicon layer **210** into slit **320** to fully or partially fill slit **320** (with or without an air gap) using one or more thin film deposition processes, such as ALD, CVD, PVD, any other suitable processes, or any combination thereof.

(71) In some embodiments, to form dielectric layer **160** (e.g., dielectric layer **334** in FIG. 3F), a silicon nitride layer is deposited over first silicon oxide layer **202** along the sidewalls and bottom surface of slit **320** (e.g., in FIGS. 2A-2D, 2G, and 2J). In one example, hydrogen or its isotope may be in situ doped into the silicon nitride layer during the deposition, such as CVD. In another example, after depositing the silicon nitride layer, the silicon nitride layer may be annealed in the presence of hydrogen or its isotope, for example, with hydrogen gas flowing through a furnace chamber. In some embodiments, the atomic percent of the doped hydrogen or its isotope in dielectric layer **160** is controlled to be greater than about 50% after the formation of dielectric layer **160**.

(72) Referring to FIGS. 2B, 2D, 2F, and 2H, in some embodiments, silicon oxide cap **208** is formed at one end of insulating structure **130**. Silicon oxide cap **208** can be formed by first etching back the top portion of at least dielectric layer **160** (and one or more of first and second silicon oxide layers **202** and **204** and polysilicon layer **210** in various examples) to form a recess, and depositing a layer of silicon oxide to fill the recess. In some embodiments, a CMP process is performed to planarize the top surface of the layer of silicon oxide to form the silicon oxide cap **208**. Referring back to FIG. 3F, an insulating structure **336** including dielectric layer **334** doped with hydrogen or its isotope is thereby formed in slit **320** (shown in FIG. 3E).

(73) Method **400** proceeds to operation **416**, as illustrated in FIG. 4, in which at least part of the at least one of hydrogen or an isotope of hydrogen is diffused to the channel structure. In some

embodiments, the at least part of the at least one of hydrogen or an isotope of hydrogen is diffused from the dielectric layer to the semiconductor channel. During any thermal processes after the formation of insulating structure **336** having hydrogen-rich dielectric layer **334**, the hydrogen or its isotope can be diffused from dielectric layer **334** through the sidewalls of insulating structure **336** to the semiconductor channel of channel structure **314** to dope the semiconductor channel, for example, polysilicon, with hydrogen or its isotope. The doped hydrogen or its isotope can form H—Si covalent bonds with the dangling bonds in the polysilicon semiconductor channel to reduce the dangling bonds therein, thereby improving the carrier mobility and adjusting the stress of channel structure **314**. The atomic percent of the hydrogen or its isotope in dielectric layer **334** can thus be reduced after diffusion, for example, to be not greater than 50%, such as between 1% and 10%. It is understood that no additional thermal process may be added to the fabrication process flow to diffuse the hydrogen or its isotope. Instead, any existing processes after the formation of insulating structure **336** in the fabrication process flow may cause the diffusion of the hydrogen or its isotope, such as the various processes described below for illustrative purposes only.

(74) As illustrated in FIG. 3G, after the formation of insulating structure **336**, local contacts, including channel local contacts **344** and word line local contacts **342**, and peripheral contacts **338** and **340** are formed. A local dielectric layer can be formed on memory stack **330** by depositing dielectric materials, such as silicon oxide or silicon nitride, using one or more thin film deposition processes, such as CVD, PVD, ALD, or any combination thereof, on top of memory stack **330**. Channel local contacts **344**, word line local contacts **342**, and peripheral contacts **338** and **340** can be formed by etching contact openings through the local dielectric layer (and any other ILD layers) using wet etching and/or dry etching, e.g., RIE, followed by filling the contact openings with conductive materials using one or more thin film deposition processes, such as ALD, CVD, PVD, any other suitable processes, or any combination thereof.

(75) As illustrated in FIG. 3H, in some embodiments, a front side source contact **343** is formed in the same processes for forming word line local contacts **342**. Front side source contact **343** can be in contact with N-type doped semiconductor layers **326**.

(76) As illustrated in FIGS. 3G and 3H, a bonding layer **346** is formed above channel local contacts **344**, word line local contacts **342**, and peripheral contacts **338** and **340** (and source contact **343** in FIG. 3H). Bonding layer **346** includes bonding contacts electrically connected to channel local contacts **344**, word line local contacts **342**, and peripheral contacts **338** and **340**. To form bonding layer **346**, an ILD layer is deposited using one or more thin film deposition processes, such as CVD, PVD, ALD, or any combination thereof, and the bonding contacts are formed through the ILD layer using wet etching and/or dry etching, e.g., RIE, followed by one or more thin film deposition processes, such as ALD, CVD, PVD, any other suitable processes, or any combination thereof.

(77) As illustrated in FIG. 3I, silicon substrate **302** and components formed thereon (e.g., memory stack **330** and channel structures **314** formed therethrough) are flipped upside down. Bonding layer **346** facing down is bonded with bonding layer **348** facing up, i.e., in a face-to-face manner, thereby forming a bonding interface **354** between silicon substrates **302** and **350**, according to some embodiments. In some embodiments, a treatment process, e.g., a plasma treatment, a wet treatment, and/or a thermal treatment, is applied to the bonding surfaces prior to the bonding. After the bonding, the bonding contacts in bonding layer **346** and the bonding contacts in bonding layer **348** are aligned and in contact with one another, such that memory stack **330** and channel structures **314** formed therethrough can be electrically connected to peripheral circuits **352** and are above peripheral circuits **352**.

(78) As illustrated in FIG. 3J, silicon substrate **302** (shown in FIG. 3I) is thinned from the backside to expose N-type doped semiconductor layer **304**. Silicon substrate **302** can be thinned using CMP, grinding, dry etching, and/or wet etching. In some embodiments, the CMP process is performed to thin silicon substrate **302** until reaching the top surface of N-type doped semiconductor layer **304**.

(79) As illustrated in FIG. 3K, one or more ILD layers **356** are formed on N-type doped semiconductor layer **304**. ILD layers **356** can be formed by depositing dielectric materials on the top surface of N-type doped semiconductor layer **304** using one or more thin film deposition processes, such as ALD, CVD, PVD, any other suitable processes, or any combination thereof. As illustrated in FIG. 3K, a source contact opening **358** is formed through ILD layers **356** into N-type doped semiconductor layer **304**. In some embodiments, source contact opening **358** is formed using wet etching and/or dry etching, such as RIE. In some embodiments, source contact opening **358** extends further into the top portion of N-type doped semiconductor layer **304**. The etching process through ILD layers **356** may continue to etch part of N-type doped semiconductor layer **304**. In some embodiments, a separate etching process is used to etch part of N-type doped semiconductor layer **304** after etching through ILD layers **356**. In some embodiments, source contact opening **358** is patterned using lithography to be aligned with insulating structure **336** at opposite sides of N-type doped semiconductor layer **304**.

(80) As illustrated in FIG. 3L, a source contact **364** is formed in source contact opening **358** (shown in FIG. 3K) at the backside of N-type doped semiconductor layer **304**. Source contact **364** is above memory stack **330** and in contact with N-type doped semiconductor layer **304**, according to some embodiments. In some embodiments, one or more conductive materials are deposited into source contact opening **358** using one or more thin film deposition processes, such as ALD, CVD, PVD, any other suitable processes, or any combination thereof, to fill source contact opening **358** with an adhesive layer (e.g., TiN) and a conductor layer (e.g., W). A planarization process, such as CMP, can then be performed to remove the excess conductive materials, such that the top surface of source contact **364** is flush with the top surface of ILD layers **356**. In some embodiments, as source contact opening **358** is aligned with insulating structure **336**, backside source contact **364** is aligned with insulating structure **336** as well.

(81) As illustrated in FIG. 3M, a redistribution layer **370** is formed above and in contact with source contact **364**. In some embodiments, redistribution layer **370** is formed by depositing a conductive material, such as Al, on the top surfaces of N-type doped semiconductor layer **304** and source contact **364** using one or more thin film deposition processes, such as ALD, CVD, PVD, any other suitable processes, or any combination thereof. As illustrated in FIG. 3M, a passivation layer **372** is formed on redistribution layer **370**. In some embodiments, passivation layer **372** is formed by depositing a dielectric material, such as silicon nitride, using one or more thin film deposition processes, such as ALD, CVD, PVD, any other suitable processes, or any combination thereof. An interconnect layer **376** including ILD layers **356**, redistribution layer **370**, and passivation layer **372** is thereby formed, according to some embodiments.

(82) As illustrated in FIG. 3K, contact openings **360** and **361** each extending through

(83) ILD layers **356** and N-type doped semiconductor layer **304** are formed. Contact openings **360** and **361** and source contact opening **358** can be formed using the same etching process to reduce the number of etching processes. In some embodiments, contact openings **360** and **361** are formed using wet etching and/or dry etching, such as RIE, through ILD layers **356** and N-type doped semiconductor layer **304**. In some embodiments, contact openings **360** and **361** are patterned using lithography to be aligned with peripheral contacts **338** and **340**, respectively. The etching of contact openings **360** and **361** can stop at the upper ends of peripheral contacts **338** and **340** to expose peripheral contacts **338** and **340**. As illustrated in FIG. 3K, a spacer **362** is formed along the sidewalls of contact openings **360** and **361** to electrically isolate N-type doped semiconductor layer **304** using one or more thin film deposition processes, such as ALD, CVD, PVD, any other suitable processes, or any combination thereof.

(84) As illustrated in FIG. 3L, contacts **366** and **368** are formed in contact openings **360** and **361**, respectively (shown in FIG. 3K) at the backside of N-type doped semiconductor layer **304**.

Contacts **366** and **368** extend vertically through ILD layers **356** and N-type doped semiconductor layer **304**, according to some embodiments. Contacts **366** and **368** and source contact **364** can be

formed using the same deposition process to reduce the number of deposition processes. In some embodiments, one or more conductive materials are deposited into contact openings **360** and **361** using one or more thin film deposition processes, such as ALD, CVD, PVD, any other suitable processes, or any combination thereof, to fill contact openings **360** and **361** with an adhesive layer (e.g., TiN) and a conductor layer (e.g., W). A planarization process, such as CMP, can then be performed to remove the excess conductive materials, such that the top surfaces of contacts **366** and **368** are flush with the top surface of ILD layers **356**. In some embodiments, as contact openings **360** and **361** are aligned with peripheral contacts **338** and **340**, respectively, contacts **366** and **368** are above and in contact with peripheral contacts **338** and **340**, respectively, as well.

(85) As illustrated in FIG. 3M, redistribution layer **370** is also formed above and in contact with contact **366**. As a result, N-type doped semiconductor layer **326** can be electrically connected to peripheral contact **338** through N-type doped semiconductor layer **304**, source contact **364**, redistribution layer **370** of interconnect layer **376**, and contact **366**. In some embodiments, N-type doped semiconductor layers **326** and **304** are electrically connected to peripheral circuits **352** through source contact **364**, interconnect layer **376**, contact **366**, peripheral contact **338** and bonding layers **346** and **348**.

(86) As illustrated in FIG. 3M, a contact pad **374** is formed above and in contact with contact **368**. In some embodiments, part of passivation layer **372** covering contact **368** is removed by wet etching and dry etching to expose part of redistribution layer **370** underneath to form contact pad **374**. As a result, contact pad **374** for pad-out can be electrically connected to peripheral circuits **352** through contact **368**, peripheral contact **340**, and bonding layers **346** and **348**.

(87) According to one aspect of the present disclosure, a 3D memory device includes a memory stack including interleaved stack conductive layers and stack dielectric layers, a semiconductor layer, a plurality of channel structures each extending vertically through the memory stack into the semiconductor layer, and an insulating structure extending vertically through the memory stack and including a dielectric layer doped with at least one of hydrogen or an isotope of hydrogen.

(88) In some embodiments, an atomic percent of the at least one of hydrogen or an isotope of hydrogen in the dielectric layer is not greater than about 50%. In some embodiments, the atomic percent of the at least one of hydrogen or an isotope of hydrogen in the dielectric layer is between about 1% and about 10%.

(89) In some embodiments, the dielectric layer includes silicon nitride.

(90) In some embodiments, the insulating structure further includes a first silicon oxide layer laterally between the dielectric layer and the stack conductive layers of the memory stack.

(91) In some embodiments, the insulating structure further includes a second silicon oxide layer. The dielectric layer is laterally between the first and second silicon oxide layers, according to some embodiments.

(92) In some embodiments, the insulating structure further includes a polysilicon layer. The dielectric layer is laterally between the first silicon oxide layer and the polysilicon layer, according to some embodiments.

(93) In some embodiments, the insulating structure further includes a silicon oxide cap at one end of the insulating structure.

(94) In some embodiments, one end of the insulating structure is flush with a respective end of the dielectric layer.

(95) In some embodiments, the semiconductor layer includes one or more doped silicon layers.

(96) In some embodiments, each of the channel structures includes a memory film and a semiconductor channel doped with the at least one of hydrogen or an isotope of hydrogen.

(97) In some embodiments, the semiconductor channel includes polysilicon.

(98) In some embodiments, the insulating structure extends laterally to separate the plurality of channel structures into a plurality of blocks.

(99) In some embodiments, the 3D memory device further includes a source contact in contact with

the semiconductor layer. The source contact and the insulating structure are on opposite sides of the semiconductor layer, according to some embodiments.

(100) In some embodiments, the 3D memory device further includes a source contact in contact with the semiconductor layer. The source contact and the insulating structure are on a same side of the semiconductor layer, according to some embodiments.

(101) According to another aspect of the present disclosure, a 3D memory device includes a memory stack including interleaved stack conductive layers and stack dielectric layers, a semiconductor layer, a plurality of channel structures each extending vertically through the memory stack into the semiconductor layer, and an insulating structure extending vertically through the memory stack and including a dielectric layer. Each of the channel structures includes a memory film and a semiconductor channel doped with at least one of hydrogen or an isotope of hydrogen.

(102) In some embodiments, the dielectric layer is doped with the at least one of hydrogen or an isotope of hydrogen.

(103) In some embodiments, an atomic percent of the at least one of hydrogen or an isotope of hydrogen in the dielectric layer is not greater than about 50%. In some embodiments, the atomic percent of the at least one of hydrogen or an isotope of hydrogen in the dielectric layer is between about 1% and about 10%.

(104) In some embodiments, the dielectric layer includes silicon nitride.

(105) In some embodiments, the insulating structure further includes a first silicon oxide layer laterally between the dielectric layer and the stack conductive layers of the memory stack.

(106) In some embodiments, the insulating structure further includes a second silicon oxide layer. The dielectric layer is laterally between the first and second silicon oxide layers, according to some embodiments.

(107) In some embodiments, the insulating structure further includes a polysilicon layer. The dielectric layer is laterally between the first silicon oxide layer and the polysilicon layer, according to some embodiments.

(108) In some embodiments, the insulating structure further includes a silicon oxide cap at one end of the insulating structure.

(109) In some embodiments, one end of the insulating structure is flush with a respective end of the dielectric layer.

(110) In some embodiments, the semiconductor layer includes one or more doped silicon layers.

(111) In some embodiments, the semiconductor channel includes polysilicon.

(112) In some embodiments, the insulating structure extends laterally to separate the plurality of channel structures into a plurality of blocks.

(113) In some embodiments, the 3D memory device further includes a source contact in contact with the semiconductor layer. The source contact and the insulating structure are on opposite sides of the semiconductor layer, according to some embodiments.

(114) In some embodiments, the 3D memory device further includes a source contact in contact with the semiconductor layer. The source contact and the insulating structure are on a same side of the semiconductor layer, according to some embodiments.

(115) According to still another aspect of the present disclosure, a method for forming a 3D memory device is disclosed. A channel structure extending vertically through a memory stack into a semiconductor layer on a substrate is formed. The memory stack includes interleaved stack conductive layers and stack dielectric layers. An insulating structure in an opening extending vertically through the memory stack and including a dielectric layer doped with at least one of hydrogen or an isotope of hydrogen is formed.

(116) In some embodiments, at least part of the at least one of hydrogen or an isotope of hydrogen is diffused to the channel structure.

(117) In some embodiments, to form the insulating structure, a first layer of silicon oxide, the

dielectric layer, and a second layer of silicon oxide are sequentially formed in the opening.

(118) In some embodiments, to form the insulating structure, a layer of silicon oxide, the dielectric layer, and a layer of polysilicon are sequentially formed in the opening.

(119) In some embodiments, the insulating structure does not extend vertically into the N-type doped semiconductor layer.

(120) In some embodiments, to form the insulating structure, a layer of silicon oxide and the dielectric layer are sequentially formed in the opening.

(121) In some embodiments, to form the insulating structure, a silicon oxide cap is formed at an upper end of the insulating structure. In some embodiments, to form the silicon oxide cap, a top portion of at least the dielectric layer is etched back to form a recess, and a layer of silicon oxide is deposited to fill the recess.

(122) In some embodiments, to form the dielectric layer, a layer of silicon nitride is deposited, and the layer of silicon nitride is in situ doped with the least one of hydrogen or an isotope of hydrogen.

(123) In some embodiments, to form the dielectric layer, a layer of silicon nitride is deposited, and after the deposition, the layer of silicon nitride is annealed in the presence of the least one of hydrogen or an isotope of hydrogen.

(124) In some embodiments, to form the channel structure, a first semiconductor layer on the substrate, a sacrificial layer on the first semiconductor layer, and a dielectric stack on the sacrificial layer are sequentially formed, the channel structure extending vertically through the dielectric stack and the sacrificial layer into the first semiconductor layer is formed, the opening extending vertically through the dielectric stack is formed, the sacrificial layer is replaced with a second semiconductor layer through the opening, and the dielectric stack is replaced with the memory stack through the opening.

(125) In some embodiments, to form the channel structure, a channel hole extending vertically through the dielectric stack and the sacrificial layer into the first semiconductor layer is formed, and a memory film and a semiconductor channel are sequentially formed along sidewalls of the channel hole.

(126) In some embodiments, to diffuse, the at least part of the at least one of hydrogen or an isotope of hydrogen is diffused from the dielectric layer to the semiconductor channel.

(127) In some embodiments, the semiconductor channel comprises polysilicon.

(128) The foregoing description of the specific embodiments will so reveal the general nature of the present disclosure that others can, by applying knowledge within the skill of the art, readily modify and/or adapt for various applications such specific embodiments, without undue experimentation, without departing from the general concept of the present disclosure. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed embodiments, based on the teaching and guidance presented herein. It is to be understood that the phraseology or terminology herein is for the purpose of description and not of limitation, such that the terminology or phraseology of the present specification is to be interpreted by the skilled artisan in light of the teachings and guidance.

(129) Embodiments of the present disclosure have been described above with the aid of functional building blocks illustrating the implementation of specified functions and relationships thereof. The boundaries of these functional building blocks have been arbitrarily defined herein for the convenience of the description. Alternate boundaries can be defined so long as the specified functions and relationships thereof are appropriately performed.

(130) The Summary and Abstract sections may set forth one or more but not all exemplary embodiments of the present disclosure as contemplated by the inventor(s), and thus, are not intended to limit the present disclosure and the appended claims in any way.

(131) The breadth and scope of the present disclosure should not be limited by any of the above-described exemplary embodiments, but should be defined only in accordance with the following claims and their equivalents.

Claims

1. A method for forming a three-dimensional (3D) memory device, comprising: forming a channel structure extending vertically through a memory stack into a semiconductor layer on a substrate, wherein the memory stack comprises interleaved stack conductive layers and stack dielectric layers separated from each other by a high-k dielectric layer; forming an insulating structure in an opening extending vertically through the memory stack and at a distance away from the channel structure, and comprising a dielectric layer doped with at least one of hydrogen or an isotope of hydrogen, wherein the insulating structure is separated from the stack dielectric layers by the high-k dielectric layer; and forming a blocking cap layer covering an upper end of the insulating structure and between two portions of the high-k dielectric layer, wherein the blocking cap layer is in direct contact with the high-k dielectric layer.
2. The method of claim 1, further comprising diffusing at least part of the at least one of hydrogen or an isotope of hydrogen to the channel structure.
3. The method of claim 1, wherein forming the insulating structure comprises sequentially forming a first layer of silicon oxide, the dielectric layer, and a second layer of silicon oxide in the opening.
4. The method of claim 1, wherein forming the insulating structure comprises sequentially forming a layer of silicon oxide, the dielectric layer, and a layer of polysilicon in the opening.
5. The method of claim 1, wherein forming the insulating structure comprises sequentially forming a layer of silicon oxide and the dielectric layer in the opening.
6. The method of claim 3, wherein forming the blocking cap layer comprises forming a silicon oxide cap at the upper end of the insulating structure and between two portions of the high-k dielectric layer for blocking a diffusing direction of the at least one of hydrogen or an isotope of hydrogen.
7. The method of claim 6, wherein forming the silicon oxide cap comprises: etching back a top portion of at least the dielectric layer to form a recess between the two portions of the high-k dielectric layer; and depositing a layer of silicon oxide to fill the recess.
8. The method of claim 3, wherein forming the dielectric layer comprises: depositing a layer of silicon nitride; and in situ doping the layer of silicon nitride with the least one of hydrogen or an isotope of hydrogen.
9. The method of claim 3, wherein forming the dielectric layer comprises: depositing a layer of silicon nitride; and after the deposition, annealing the layer of silicon nitride in a presence of the least one of hydrogen or an isotope of hydrogen.
10. The method of claim 2, wherein forming the channel structure comprises: sequentially forming a first semiconductor layer on the substrate, a sacrificial layer on the first semiconductor layer, and a dielectric stack on the sacrificial layer; forming the channel structure extending vertically through the dielectric stack and the sacrificial layer into the first semiconductor layer; forming the opening extending vertically through the dielectric stack; replacing the sacrificial layer with a second semiconductor layer through the opening; and replacing the dielectric stack with the memory stack through the opening.
11. The method of claim 10, wherein forming the channel structure comprises: forming a channel hole extending vertically through the dielectric stack and the sacrificial layer into the first semiconductor layer; and sequentially forming a memory film and a semiconductor channel along sidewalls of the channel hole.
12. The method of claim 11, wherein the diffusing comprises diffusing the at least part of the at least one of hydrogen or an isotope of hydrogen from the dielectric layer to the semiconductor channel.
13. The method of claim 11, wherein the semiconductor channel comprises polysilicon.
14. The method of claim 1, wherein the dielectric layer comprises silicon nitride.

15. The method of claim 1, wherein the semiconductor layer comprises one or more doped silicon layers.
16. The method of claim 1, wherein the insulating structure extends laterally to separate a plurality of channel structures into a plurality of blocks.
17. The method of claim 1, further comprising forming a plurality of channel local contact connected to a plurality of channel structures, respectively; and forming a first bonding layer above the plurality of channel local contacts.
18. The method of claim 17, further comprising forming a second substrate; forming a peripheral circuit on the second substrate; and forming a second bonding layer on the peripheral circuit.
19. The method of claim 18, further comprising bonding the first bonding layer with the second bonding layer.
20. The method of claim 1, further comprising removing the substrate to expose the semiconductor layer; forming one or more insulating layer above the exposed semiconductor layer; forming a source contact opening extending into the semiconductor layer; and forming a source contact in the source contact opening, wherein the source contact and the insulating structure are on opposite sides of the semiconductor layer.
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